

IGCSE Chemistry CIE

YOUR NOTES



9. Metals

CONTENTS

9.1 Properties, Uses & Alloys of Metals

9.1.1 Properties of Metals

9.1.2 Uses of Metals

9.1.3 Alloys

9.2 Reactivity Series & Corrosion of Metals

9.2.1 Reactivity Series

9.2.2 Explaining Reactivity

9.2.3 Rusting of Iron

9.2.4 Galvanising & Sacrificial Protection

9.3 Extraction of Metals

9.3.1 Extraction of Metals

9.3.2 Extraction of Iron from Hematite

9.3.3 Extraction of Aluminium from Bauxite

9.1 Properties, Uses & Alloys of Metals

9.1.1 Properties of Metals

Physical Properties of Metals & Non-Metals

Metals and non-metals

- The Periodic Table contains over 100 different elements
- They can be divided into two broad types: **metals** and **non-metals**
- Most of the elements are metals and a small number of elements display properties of both types
 - These elements are called **metalloids** or **semimetals**

BASIC OXIDES

ACIDIC OXIDES

1/I	2/II											3/III	4/IV	5/V	6/VI	7/VII	0/VIII
Li	Be											B	C	N	O	F	He
Na	Mg											Al	Si	P	S	Cl	Ar
K	Ca	Sc	Ti	V	Cr	Mn	Fe	Co	Ni	Cu	Zn	Ga	Ge	As	Se	Br	Kr
Rb	Sr	Y	Zr	Nb	Mo	Tc	Ru	Rh	Pd	Ag	Cd	In	Sn	Sb	Te	I	Xe
Cs	Ba	La	Hf	Ta	W	Re	Os	Ir	Pt	Au	Hg	Tl	Pb	Bi	Po	At	Rn
Fr	Ra	Ac															

Ce	Pr	Nd	Pm	Sm	Eu	Gd	Tb	Dy	Ho	Er	Tm	Yb	Lu
Th	Pa	U	Np	Pu	Am	Cm	Bk	Cf	Es	Fm	Md	No	Lr

KEY

METAL

METALLOID

NONMETAL



The metallic character diminishes moving left to right across the Periodic Table

Properties of metals

- Conduct **heat** and **electricity**
- Are **malleable** (can be hammered and made into different shapes) and **ductile** (can be drawn into wires)

YOUR NOTES



- Tend to be **lustrous** (shiny)
- Have **high** density and usually have **high** melting points
- Form **positive** ions through electron loss
- Form **basic oxides**

Properties of non-metal elements

- Do **not** conduct heat and electricity
- Are **brittle** when solid and easily break up
- Tend to be dull and nonreflective
- Have **low** density and **low** melting points (many are gases at room temperature)
- Form **negative** ions through electron gain (except for hydrogen)
- Form **acidic oxides**

YOUR NOTES



Chemical Properties of Metals

YOUR NOTES



General chemical properties of metals

- The chemistry of metals is studied by analysing their reactions with water, dilute acid and oxygen
- Based on these reactions, a reactivity series of metals can be produced

Reactivity with water

- Some metals react with water, either warm or cold, or with steam
- Metals that react with cold water form a metal hydroxide and hydrogen gas



- For example calcium:

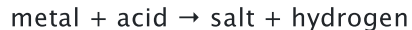


- Metals that react with steam form metal oxide and hydrogen gas, for example zinc:



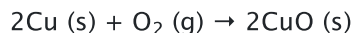
Reactivity with acids

- Most metals react with dilute acids such as HCl
- When acids and metals react, the hydrogen atom in the acid is replaced by the metal atom to produce a salt and hydrogen gas, for example iron:



Reactivity with oxygen

- Unreactive metals such as gold and platinum do not react with oxygen
- Some reactive metals such as the alkali metals react easily with oxygen
- Copper and iron can also react with oxygen although much more slowly
- When metals react with oxygen a metal oxide is formed, for example copper:



9.1.2 Uses of Metals

YOUR NOTES

**Uses of Metals****Uses of Aluminium**

Use	Most important property
Aeroplane bodies	High strength-to-weight ratio, low density
Overhead power cables	Good conductor of electricity, low density
Saucepans	Good conductor of heat
Food cans	Non-toxic, resistant to corrosion and acidic food stuffs
Window frames	Resistant to corrosion

Copyright © Save My Exams. All Rights Reserved

Uses of Copper

USE	MOST IMPORTANT PROPERTY
ELECTRICAL WIRES	GOOD CONDUCTOR OF ELECTRICITY AND MALLEABLE
WATER PIPES	EASY TO WORK WITH AND BEND, NON-TOXIC AND UNREACTIVE (DOES NOT REACT WITH WATER)

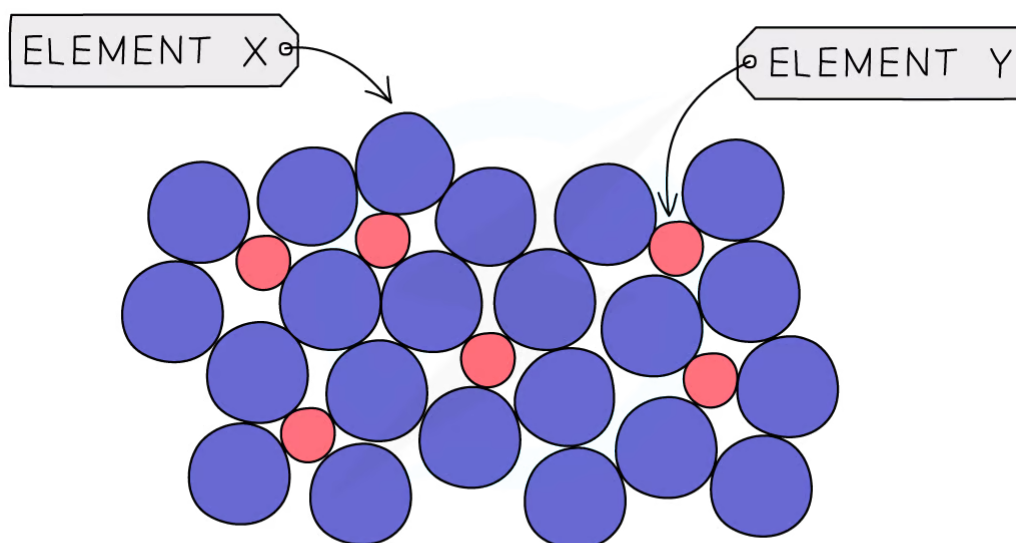
9.1.3 Alloys

YOUR NOTES



Properties & Uses of Alloys

- An alloy is a **mixture** of two or more metals or metal with a non-metal such as carbon
- Alloys often have **properties** that can be very **different** from the metals they contain, for example, they can have more **strength**, **hardness** or **resistance to corrosion** or extreme **temperatures**
- These enhanced properties can make alloys more useful than pure metals



Copyright © Save My Exams. All Rights Reserved



The regular arrangement of a metal lattice structure is distorted in alloys

Common alloys and their uses

- **Brass** is an alloy of copper and zinc and is much **stronger** than either metal
 - It is used in musical instruments, ornaments and door knobs
- **Stainless steel** is a mixture of iron and other elements, for example, chromium, nickel and carbon
 - It is used in cutlery because of its hardness and resistance to corrosion
- Alloys of iron with tungsten are extremely **hard** and resistant to high **temperatures**
- Alloys of iron mixed with chromium or nickel are resistant to **corrosion**
- Aluminium is mixed with copper, manganese and silicon for aircraft body production as the alloy is **stronger** but still has a low **density**



Exam Tip

Alloys are mixtures of substances, they are not chemically combined and an alloy is not a compound.

YOUR NOTES



Explaining the Properties of Alloys

EXTENDED

- Alloys contain atoms of **different** sizes, which **distorts** the normally regular arrangements of atoms in metals
- This makes it more difficult for the layers to **slide** over each other, so alloys are usually much harder than the pure metal

9.2 Reactivity Series & Corrosion of Metals

9.2.1 Reactivity Series

Reactivity Series

- The chemistry of the metals is studied by analysing their reactions with water and acids
- Based on these reactions a reactivity series of metals can be produced
- The series can be used to place a group of metals in **order of reactivity** based on the observations of their reactions with water and acids
- The non-metals hydrogen and carbon are also included in the reactivity series as they are used to extract metals from their oxides

Table of Metal Reactions

YOUR NOTES



YOUR NOTES



Metal	Reaction with cold water	Reaction with acid	Reaction with oxygen
Most reactive			
Potassium	Reacts violently	Reacts violently	Reacts quickly in air
Sodium	Reacts quickly	Reacts violently	Reacts quickly in air
Calcium	Reacts less strongly	Reacts vigorously	Reacts readily
Magnesium	Slow reaction (reacts with steam)	Reacts vigorously	Reacts readily
Aluminium	Slow reaction (reacts with steam)	Reacts readily	Reacts readily
Carbon			
Zinc	Very slow reaction (reacts slowly with steam)	Reacts less strongly	Reacts
Iron	Very slow reaction (reacts slowly with steam)	Reacts less strongly	Reacts
Hydrogen			
Copper	No reaction with water or steam	No reaction	Reacts
Silver	No reaction with water or steam	No reaction	Reacts
Gold	No reaction with water or steam	No reaction	No reaction
Least reactive			

Copyright © Save My Exams. All Rights Reserved

The reactivity series mnemonic

- Observations from the table above allow the following reactivity series to be deduced
- The order of this reactivity series can be memorised using the following mnemonic
 - "Please send cats, monkeys and cute zebras into hot countries signed Gordon"

YOUR NOTES



Metal	Abbreviation
Most reactive	
Potassium	P – Please
Sodium	S – Send
Calcium	C – Cats,
Magnesium	M – Monkeys,
Aluminium	A – And
Carbon	C – Cute
Zinc	Z – Zebras
Iron	I – Into
Hydrogen	H – Hot
Copper	C – Countries
Silver	S – Signed
Gold	G – Gordon
Least reactive	

Copyright © Save My Exams. All Rights Reserved

You can learn the reactivity series with the help of a silly phrase

Reactions of Metals

Reaction with cold water

- The more reactive metals will react with cold water to form a metal hydroxide and hydrogen gas
- Potassium, sodium and calcium all undergo reactions with cold water as they are the most reactive metals:



- For example, calcium and potassium:



Reaction with steam



- Metals just below calcium in the reactivity series do not react with cold water but will react with steam to form a metal oxide and hydrogen gas, for example, magnesium:



Reaction with dilute acids

- Only metals **above hydrogen** in the reactivity series will react with dilute acids
- Unreactive metals below hydrogen, such as gold, silver and copper, do not react with acids
- The more reactive the metal then the more vigorous the reaction will be
- Metals that are placed high on the reactivity series such as potassium and sodium are very dangerous and react **explosively** with acids
- When acids react with metals they form a **salt** and **hydrogen gas**:
- The general equation is:



- Some examples of metal-acid reactions and their equations are given below:

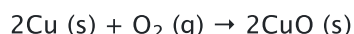
Acid-Metal Reactions Table

Acid	Sulfuric Acid	Hydrochloric Acid
Magnesium	$\text{Mg} + \text{H}_2\text{SO}_4 \rightarrow \text{MgSO}_4 + \text{H}_2$	$\text{Mg} + 2\text{HCl} \rightarrow \text{MgCl}_2 + \text{H}_2$
Zinc	$\text{Zn} + \text{H}_2\text{SO}_4 \rightarrow \text{ZnSO}_4 + \text{H}_2$	$\text{Zn} + 2\text{HCl} \rightarrow \text{ZnCl}_2 + \text{H}_2$
Iron	$\text{Fe} + \text{H}_2\text{SO}_4 \rightarrow \text{FeSO}_4 + \text{H}_2$	$\text{Fe} + 2\text{HCl} \rightarrow \text{FeCl}_2 + \text{H}_2$

Copyright © Save My Exams. All Rights Reserved

Reaction with oxygen

- Some reactive metals, such as the alkali metals, react easily with oxygen
- Silver, copper and iron can also react with oxygen although much more slowly
- When metals react with oxygen a metal oxide is formed, for example, copper:



- Gold does not react with oxygen

Deducing the order of reactivity

- The order of reactivity of metals can be deduced by making experimental observations of reactions between metals and water, acids and oxygen
- The more vigorous the reaction of the metal, the higher up the reactivity series the metal is

- A combination of reactions may be needed, for example, the order of reactivity of the more reactive metals can be determined by their reactions with water
- The less reactive metals react slowly or not at all with water, so the order of reactivity would need to be determined by observing their reactions with dilute acid
- Temperature change in a reaction can also be used to determine the order of reactivity
- The greater the temperature change in a reaction involving a metal, the more reactive the metal is

YOUR NOTES



9.2.2 Explaining Reactivity

YOUR NOTES



Explaining Reactivity

EXTENDED

- Metal atoms form **positive ions** by loss of electrons when they react with other substances
- The **tendency** of a metal to lose electrons is a measure of how reactive the metal is
- A metal that is high up on the series loses electrons easily and is thus **more reactive** than one which is lower down on the series

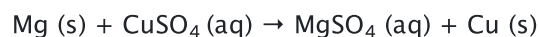
Displacement reactions between metals and aqueous solutions of metal salts

- Any metal will displace another metal that is **below** it in the reactivity series from a solution of one of its salts
- This is because more reactive metals lose electrons and form ions more readily than less reactive metals, making them better **reducing agents**
- The less reactive metal is a better electron acceptor than the more reactive metal, thus the less reactive metal is reduced. (OIL-RIG: reduction is gain of electrons)

Example: Magnesium + copper sulfate

- Magnesium is a reactive metal and can displace copper from a copper sulfate solution
- Magnesium loses its electrons more easily and the ion of the less reactive metal, copper, will gain these electrons to form elemental copper
- This is easily seen as the more reactive metal slowly **disappears** from the solution, **displacing** the less reactive metal

magnesium + copper sulfate → magnesium sulfate + copper



- The blue colour of the CuSO_4 solution **fades** as colourless magnesium sulfate solution is formed
- Copper coats the surface of the magnesium and also forms solid metal which falls to the bottom of the beaker

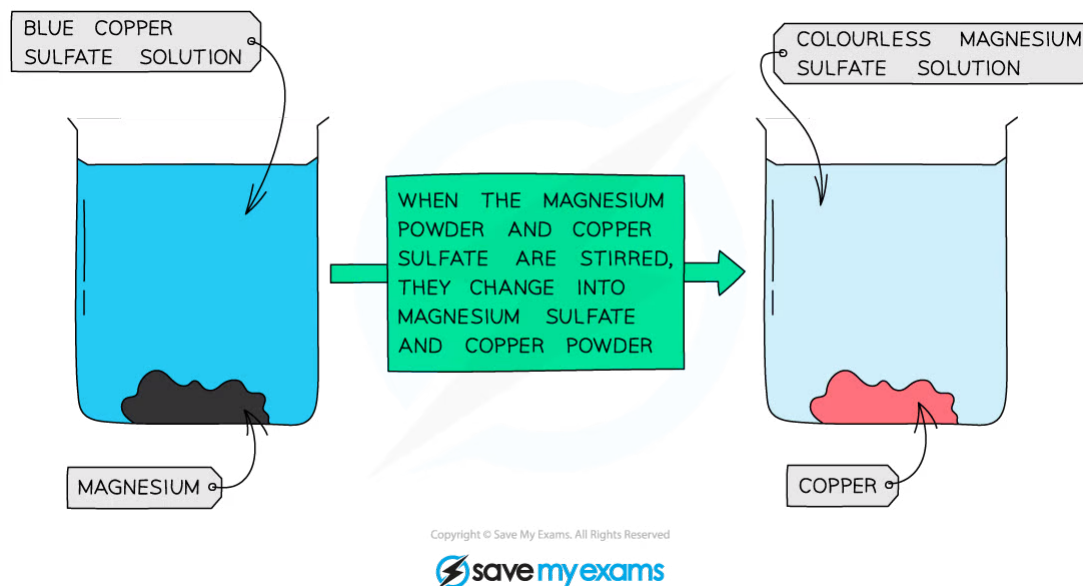


Diagram showing the colour change when magnesium displaces copper from copper sulfate

- By combining different metals and metal salts solutions it is possible to come up with a relative reactivity order

Metal Solutions Displacement Table

Mixture	Products	Equation for reaction
Magnesium and iron(II) sulfate	Magnesium sulfate and iron	$\text{Mg} + \text{FeSO}_4 \longrightarrow \text{MgSO}_4 + \text{Fe}$
Zinc and iron chloride	Zinc chloride and iron	$\text{Zn} + \text{FeCl}_2 \longrightarrow \text{ZnCl}_2 + \text{Fe}$
Iron and silver nitrate	Iron(III) nitrate and silver	$\text{Fe} + 2\text{AgNO}_3 \longrightarrow \text{Fe}(\text{NO}_3)_2 + 2\text{Ag}$
Copper and iron(II) chloride	No reaction as iron is more reactive than copper	—
Silver and copper(II) sulfate	No reaction as copper is more reactive than silver	—
Zinc and magnesium chloride	No reaction as magnesium is more reactive than zinc	—

Copyright © Save My Exams. All Rights Reserved

- From this table we can deduce the order of reactivity:
 - Magnesium and zinc are both more reactive than iron but magnesium is more reactive than zinc
 - Copper and silver are both less reactive than iron but silver is less reactive than copper

YOUR NOTES



- The order of reactivity of the metals tested can be therefore be deduced as:
 - $\text{Mg} > \text{Zn} > \text{Fe} > \text{Cu} > \text{Ag}$

Reactivity of aluminium

- Aluminium is high in the reactivity series, but in reality, it does not react with water and the reaction with dilute acids can be quite slow
- This is because it reacts readily with oxygen, forming a protective layer of aluminium oxide which is very thin
- This layer prevents reaction with water and dilute acids, so aluminium can behave as if it is unreactive

YOUR NOTES



9.2.3 Rusting of Iron

YOUR NOTES



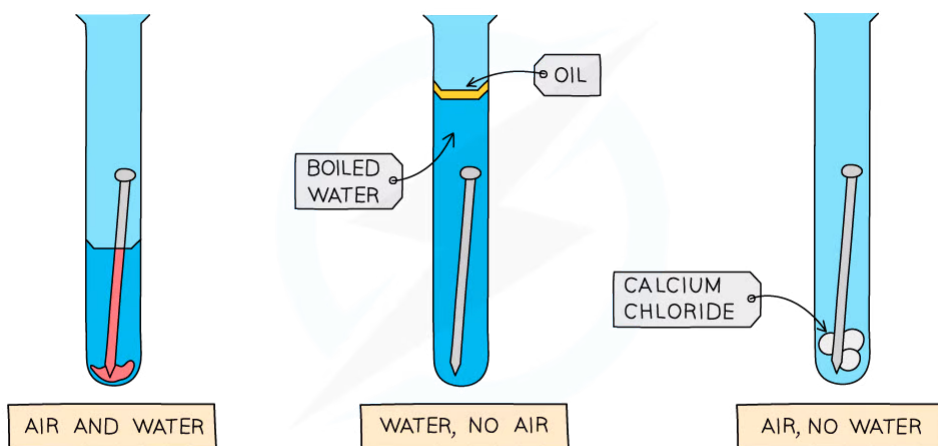
Rusting of Iron

- Rust is a chemical reaction between iron, water and oxygen to form the compound hydrated iron(III) oxide (rust)
- Oxygen and water must be present for rust to occur
- During rusting, iron is oxidised



Investigating rusting

- To investigate the conditions required for rusting, prepare three test tubes as shown in the diagram
- The oil in the 2nd tube keeps out air and the water has been boiled so that no air is left in it
- The calcium chloride in the 3rd tube is used to remove any moisture in the air
- After a few days, the iron nail in the 1st tube will be the only nail to show signs of rust



Copyright © Save My Exams. All Rights Reserved

Diagram showing the requirements of oxygen and water for rust to occur: only the nail on the left rusts

Rust prevention methods

Barrier methods

- Rust can be prevented by coating iron with barriers that prevent the iron from coming into contact with water and oxygen
- However, if the coatings are washed away or scratched, the iron is once again exposed to water and oxygen and will rust

COMMON BARRIER METHODS	
GREASE	OIL
PAINT	PLASTIC

**Exam Tip**

Only iron or steel (an alloy made from iron) can **rust**. If any other metal oxidises in air causing the metal to break down, you should say that the metal has **corroded**.

YOUR NOTES



9.2.4 Galvanising & Sacrificial Protection

YOUR NOTES



Galvanising & Sacrificial Protection

EXTENDED

- Iron can be prevented from rusting using the **reactivity series**

Sacrificial Protection

- A **more** reactive metal can be attached to a **less** reactive metal
- The more reactive metal will oxidise and therefore corrode first, protecting the less reactive metal from corrosion
- E.g. using zinc bars on the side of steel ships:

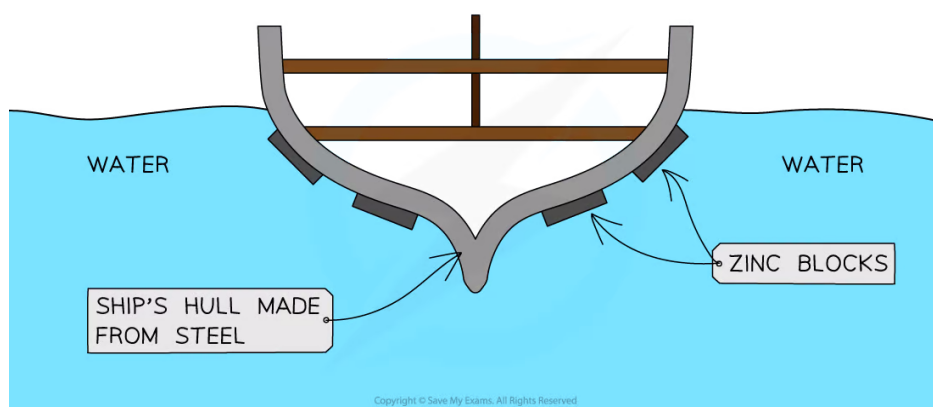
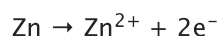


Diagram to show the use of zinc bars on the sides of steel ships as a method of sacrificial protection

- Zinc is more reactive than iron therefore will lose its electrons more easily than iron and is oxidised more easily:



- The iron is less reactive therefore will not lose its electrons as easily so it is not oxidised; the zinc is **sacrificed** to protect the steel
- For continued protection, the zinc bars have to be replaced before they completely corrode

Galvanising

- Galvanising** is a process where the iron to be protected is coated with a layer of zinc
- This can be done by electroplating or dipping it into molten zinc
- ZnCO_3 is formed when zinc reacts with oxygen and carbon dioxide in the air and protects the iron by the barrier method
- If the coating is damaged or scratched, the iron is still protected from rusting by **sacrificial protection**



Exam Tip

You may be asked to explain why a metal is/is not suitable as a method of preventing an iron/steel object from rusting. Remember that if it is higher in the reactivity series than iron, it will be suitable for sacrificial protection as it will be oxidised instead of iron. If it is lower in the reactivity series than iron, it would not be suitable as iron would be oxidised, causing it to rust.

YOUR NOTES



9.3 Extraction of Metals

9.3.1 Extraction of Metals

Extraction of Metals

- The Earth's crust contains metals and metal compounds such as gold, copper, iron oxide and aluminium oxide
- Useful metals are often **chemically combined** with other substances forming **ores**
- A metal ore is a rock that contains enough of the metal to make it worthwhile extracting
- They have to be extracted from their ores through processes such as **electrolysis**, using a **blast furnace** or by reacting with **more reactive** material
- In many cases the ore is an **oxide** of the metal, therefore the extraction of these metals is a reduction process since oxygen is being removed
- Common examples of oxide ores are **iron** and **aluminium** ores which are called **hematite** and **bauxite** respectively
- Unreactive metals do not have to be extracted chemically as they are often found as the **uncombined** element
- This occurs as they do not easily react with other substances due to their **chemical stability**
- They are known as **native metals** and examples include **gold** and **platinum** which can both be mined directly from the Earth's crust
- The position of the metal on the reactivity series influences the method of extraction
- Those metals placed higher up on the series (above carbon) have to be extracted using electrolysis
- Metals lower down on the series can be extracted by heating with carbon

The Extraction Method Depends on the Position of a Metal in the Reactivity Series

YOUR NOTES



YOUR NOTES



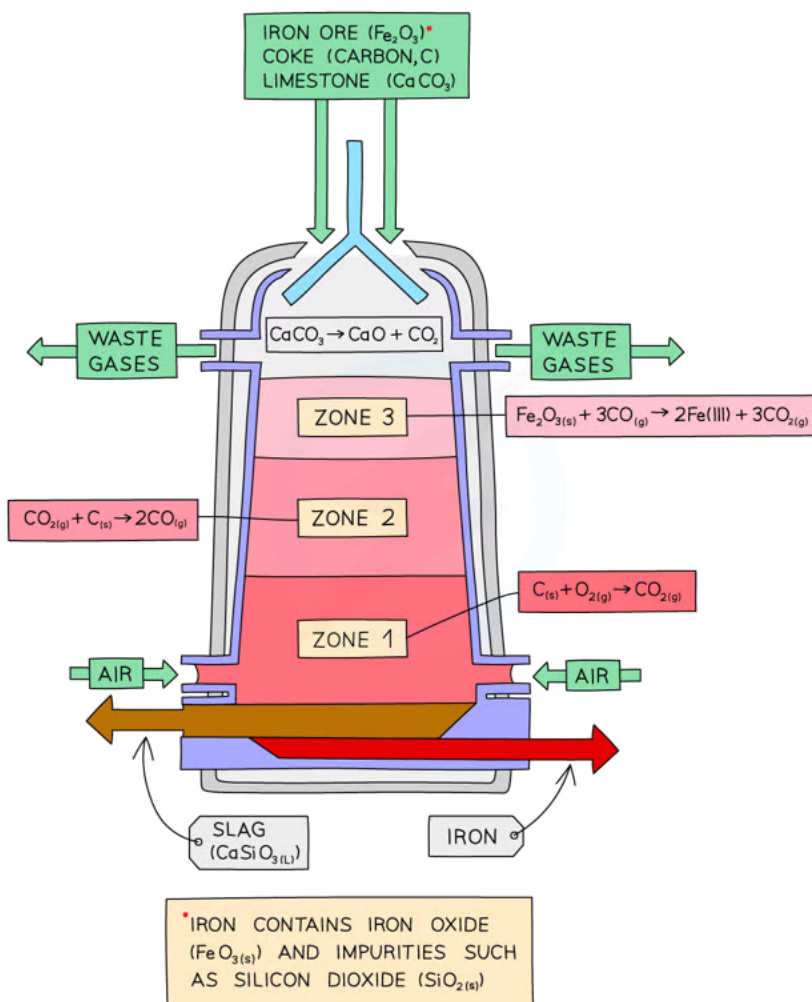
Metal	Method of extraction
Most reactive	
Potassium	Extracted by electrolysis of the molten chloride or molten oxide Large amount of electricity required so expensive process
Sodium	
Lithium	
Calcium	
Magnesium	
Aluminium	
Carbon	
Zinc	Extracted by heating with a reducing agent such as carbon or carbon monoxide in a blast furnace Cheap process as carbon is cheap and can be source of heat as well Found as pure elements
Iron	
Hydrogen	
Copper	
Silver	
Gold	
Least reactive	

Copyright © Save My Exams. All Rights Reserved

9.3.2 Extraction of Iron from Hematite

Extraction of Iron from Hematite

- Iron is extracted in a large container called a blast furnace from its ore, hematite
- Modern blast furnaces produce approximately 10,000 tonnes of iron per day
- The process is demonstrated and explained below:



Copyright © Save My Exams. All Rights Reserved

Diagram showing the carbon extraction of iron

- The raw materials: iron ore (hematite), coke (an impure form of carbon), and limestone are added into the top of the blast furnace
- Hot air is blown into the bottom
- **Zone 1:**
 - Coke **burns** in the hot air forming **carbon dioxide**
 - The reaction is exothermic so it gives off heat, **heating** the furnace

YOUR NOTES



carbon + oxygen → carbon dioxide

- **Zone 2:**

- At the high temperatures in the furnace, more coke reacts with carbon dioxide forming carbon monoxide
- Carbon dioxide has been **reduced** to carbon monoxide

carbon + carbon dioxide → carbon monoxide

- **Zone 3:**

- Carbon monoxide **reduces** the iron(III) oxide in the iron ore to form iron
- This will melt and collect at the bottom of the furnace, where it is tapped off:

iron(III) oxide + carbon monoxide → iron + carbon dioxide

- Limestone (calcium carbonate) is added to the furnace to remove impurities in the ore.
 - The calcium carbonate in the limestone **thermally decomposes** to form calcium oxide

calcium carbonate → calcium oxide + carbon dioxide

- The calcium oxide formed reacts with the silicon dioxide, which is an impurity in the iron ore, to form calcium silicate
- This melts and collects as a molten **slag** floating on top of the molten iron, which is tapped off separately

calcium oxide + silicon dioxide → calcium silicate



Exam Tip

For Core students, the symbol equations are **not** needed for the different reactions involved in the extraction of iron from hematite.

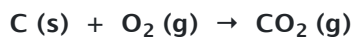
YOUR NOTES



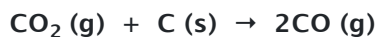
Equations for Extraction of Iron from Hematite

EXTENDED

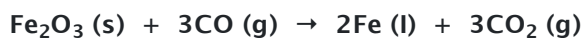
- The symbol equations for the different stages of the extraction of iron from hematite are:
- Zone 1:** The burning of carbon (coke) to provide heat and produce carbon dioxide:



- Zone 2:** The reduction of carbon dioxide to carbon monoxide:



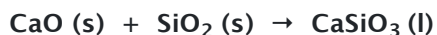
- Zone 3:** The reduction of iron(III) oxide by carbon monoxide:



- The thermal decomposition of calcium carbonate (limestone) to produce calcium oxide:



- The formation of slag:



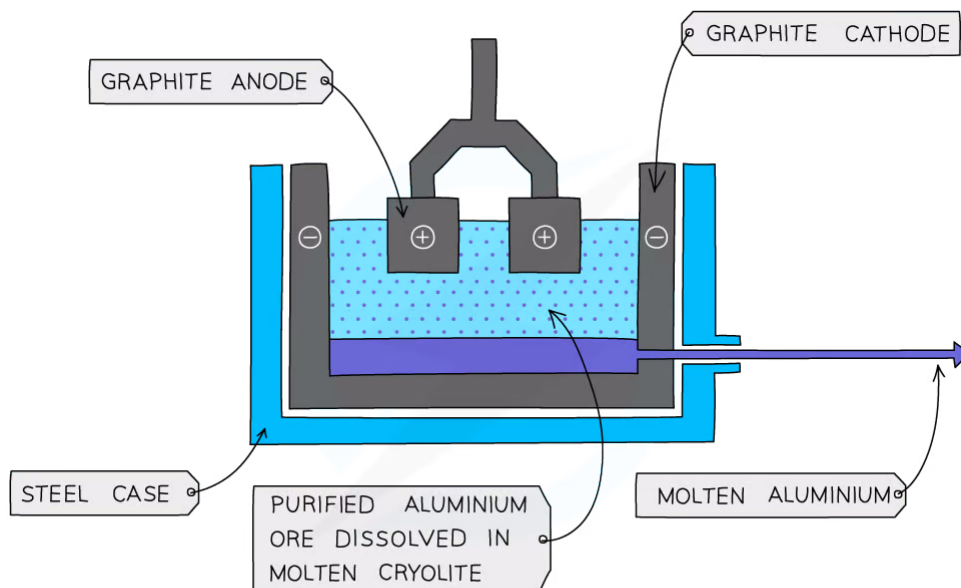
YOUR NOTES



9.3.3 Extraction of Aluminium from Bauxite

Extraction of Aluminium from Bauxite

- Aluminium is a reactive metal, above carbon in the reactivity series
- Its main ore, is **bauxite**, which contains aluminium oxide
- Aluminium is higher in the reactivity series than carbon, so it cannot be extracted by reduction using carbon
- Instead, aluminium is extracted by **electrolysis**



Copyright © Save My Exams. All Rights Reserved

Diagram showing the extraction of aluminium by electrolysis



Exam Tip

If you are a Core student, you do **not** need to explain the process of extraction of aluminium by electrolysis.

YOUR NOTES



The Process of Aluminium Extraction by Electrolysis

EXTENDED

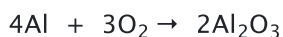
- Bauxite is first **purified** to produce aluminium oxide, Al_2O_3
- Aluminium oxide is then dissolved in **molten cryolite**
 - This is because aluminium oxide has a melting point of over 2000°C which would use a lot of energy and be very expensive
 - The resulting mixture has a lower melting point without interfering with the reaction
- The mixture is placed in an electrolysis cell, made from steel, lined with graphite
- The **graphite** lining acts as the negative electrode, with several large graphite blocks as the positive electrodes
- At the **cathode** (negative electrode):
 - Aluminium ions gain electrons (reduction)
 - Molten aluminium forms at the bottom of the cell
 - The molten aluminium is siphoned off from time to time and fresh aluminium oxide is added to the cell



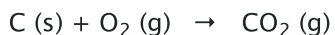
- At the **anode** (positive electrode):
 - Oxide ions lose electrons (oxidation)
 - Oxygen is produced at the anode:



- The overall equation for the reaction is:



- The carbon in the **graphite** anodes reacts with the oxygen produced to produce **CO_2**

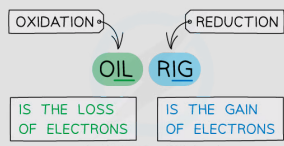


- As a result the anode wears away and has to be replaced regularly
- A lot of **electricity** is required for this process of extraction, this is a major **expense**



Exam Tip

Use OIL RIG to remember whether oxidation or reduction has occurred at the electrodes:



YOUR NOTES



